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CORFLEX® MC

Continuous Corrugated Aluminum Sheathed PL, MC & MV Industrial Cables

Description



Standards

National UL 1569; UL 2225; UL 44

Applications

CORFLEX® cables may be found in a variety of industrial, commercial and utility applications. Chemical, oil and gas, and forestry industries plus commercial or high-rise buildings are areas in which CORFLEX® has gained acceptance.

Exceptional fire ratings (as per appropriate specifications), impact-resistance, flexiblity and an impervious metallic sheath are key components of this design.

CORFLEX® installation may be in wet or dry locations, in trays, troughs, wireways, directly buried or embedded in concrete. As well, it can be used in plenums (with no outer coverings), ducts and other airways per NEC 2008 and NEC 2011 Article 300.22.

The 300 volt CORFLEX® PL cables can be installed in Class I, II, III & intrinsically safe locations & Division 2 as per NEC 2008 and NEC 2011 (PLTC and ITC Rated).

CORFLEX® PL cables are suitable for control, signal, and instrumentation circuits with 300 volt rating. The cable construction minimizes noise and signal interference to enhance performance in instrumentation, dataflow, computer & datalogging applications especially in areas where high voltages or high currents are present.

The 600 volt and higher CORFLEX® cables can be installed in hazardous locations designated Class I, II & III and Divisions 1 & 2 as per NEC 2008 and NEC 2011 (HL Rated).

CORFLEX® MC cables are designed for use on power and control circuits with 600 volt rating. Cables are suitable for use as feeders and branch circuits for power, control, lighting and signalling as per NEC 2008 and NEC 2011 Articles 330, 725 & 727.

CORFLEX® MV-105 or**MC-HL** cables are designed for use on power circuits. Cables are suitable for use as per NEC 2008 and NEC 2011 Article 328 as feeders and branch circuits.

CORFLEX® VFD cables are designed to improve the operating performance of variable frequency drive systems. The 600 volt CORFLEX® VFD cables provide excellent shielding from high frequency noise that can interfere with data and control signals. One of the advantages of the Nexans CORFLEX® sheath compared to other constructions is that the cable sheath provides a long term low resistance path to ground to protect the drive system. The cable grounds provide the best cancellation and lowest net injected ground current into the drive system, protecting against common mode currents which impair the drive unit electronic performance and shorten the drive's life span. Cable grounds, in combination with the aluminum sheath, ensure a balanced, very low resistance path to ground to reduce the chance of motor failure due to bearing currents.

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CORFLEX® MC

Armored Instrumentation Cable

Part Number: MC

Armored Instrumentation Cable UL Type MC HL 600V, 90°C rated - LEAD FREE

Description

Single or multiple individually shielded pairs or triads, overall cable shield, PVC inner jacket, continuous corrugated aluminum sheath, PVC jacket



Standards

National UL 1569;UL 2225;UL 66

Construction

Conductor: Bare, annealed copper conforming to ASTM B3 and Class B stranded in accordance with ASTM B8.

Insulation: PVC/Nylon type TFN in accordance with UL 66, flame retardant, 90oC temperature rating.

Insulation shield: Aluminum foil/polyester shield helically wrapped to provide 100% coverage and tinned copper drain wire that is two gauge sizes smaller than the circuit conductors.

Assembly: Pairs/triads are cabled in concentric layers with interstices filled with suitable non-hygroscopic fillers, as required. A binder tape of synthetic material assembles the core in an essentially round configuration.

Overall cable shield: Aluminum foil/polyester shield helically wrapped to provide 100% coverage and tinned copper drain wire that is the same size as the circuit conductors.

Inner jacket: Polyvinyl chloride jacket, over cabled core as per UL 1569, 90°C temperature rating, with additional resistance to fire spread. A rip cord is laid longitudinally under the jacket to facilitate stripping.

Armor: Continuous corrugated aluminum sheath with no more than 0.2% trace copper providing complete protection against liquid & gas ingress. It also provides excellent mechanical protection, additional electrostatic shielding, and serves as an easy means of grounding equipment.

Jacket: Overall black polyvinyl chloride jacket per UL 1569, 90°C temperature rating; low gas emission; limited flame spread and excellent corrosion resistance.

Identification of Conductors

Pairs: black/white & number coded

Triads: black/white/red & number coded

Bending Radius

Fixed position: 7 x cable overall diameter

During pulling: 12 x cable overall diameter

Specifications
* Meets UL 66, TFN rated 90°C 600V conductors
* Meets UL 1569 requirements for Type MC, Metal Clad cables
* Meets UL 2225 for Hazardous Locations
* Designated Type MC as per NEC Article 330

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Characteristics

Construction characteristics	
Conductor material	Copper
Electrical characteristics	
Maximum operating voltage	600 V
Usage characteristics	
Maximum operating temperature	90 °C

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Multi Pairs, 600 V - 18 AWG (7w) SPOS

Part Number	# of Pairs	Insulation Thickness		Nominal Diameter over Core (inches)	Inner Jacket Thickness (mils)	Nominal Diameter over Inner Jacket (inches)	Nominal Diameter over Sheath (inches)	Outer Jacket Thickness (mils)	Nominal Diameter over Outer Jacket (inches)	Approximate Net Cable Weight (lb/kft)
		PVC (mils)	Nylon (mils)							
654988	2	15	4	0.319	40	0.405	0.594	50	0.699	218
654921	4	15	4	0.38	40	0.465	0.633	50	0.735	257
654947	8	15	4	0.517	50	0.621	0.836	50	0.940	373
654962	12	15	4	0.647	50	0.753	0.976	50	1.080	621
665250	16	15	4	0.731	50	0.837	1.142	50	1.247	817
662116	24	15	4	1.016	50	1.124	1.420	50	1.525	969

Multi Pairs, 600 V - 16 AWG (7w) SPOS

Part Number	# of Pairs	Insulation Thickness		Nominal Diameter over Core (inches)	Inner Jacket Thickness (mils)	Nominal Diameter over Inner Jacket (inches)	Nominal Diameter over Sheath (inches)	Outer Jacket Thickness (mils)	Nominal Diameter over Outer Jacket (inches)	Approximate Net Cable Weight (lb/kft)
		PVC (mils)	Nylon (mils)							
*645390	1	15	4	0.209	40	0.293	0.492	50	0.605	157
*659052	2	15	4	0.424	40	0.477	0.639	50	0.744	216
*645291	4	15	4	0.482	50	0.533	0.767	50	0.871	375
*659078	8	15	4	0.635	50	0.676	0.926	50	1.030	604
*650796	12	15	4	0.797	50	0.835	1.140	50	1.246	862
654889	16	15	4	0.882	50	0.983	1.320	50	1.431	1151
*654905	24	15	4	1.124	50	1.126	1.422	50	1.526	1432
650788	36	15	4	1.312	50	1.370	1.746	60	1.879	2121
* Stock Items										

Multi Triads, 600 V - 16 AWG (7w) STOS

Part Number	# of Pairs	Insulation Thickness		Nominal Diameter over Core (inches)	Inner Jacket Thickness (mils)	Nominal Diameter over Inner Jacket (inches)	Nominal Diameter over Sheath (inches)	Outer Jacket Thickness (mils)	Nominal Diameter over Outer Jacket (inches)	Approximate Net Cable Weight (lb/kft)
		PVC (mils)	Nylon (mils)							
*654863	1	15	4	0.225	40	0.304	0.507	50	0.619	171
644344	4	15	4	0.484	50	0.587	0.798	50	0.900	300
670083	8	15	4	0.680	50	0.786	0.998	50	1.102	467
*670067	12	15	4	0.820	50	0.925	1.212	50	1.317	1007
* Stock items										

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Electrical Properties

600 V Shielded Pairs / Triads with an overall Cable Shield

Conductor Size (AWG)	DC Resistance 20°C Ω/kft	Capacitance			
		Pairs		Triads	
		Conductor -Conductor (pf/ft)	Conductor -Shield (pf/ft)	Conductor -Conductor (pf/ft)	Conductor -Shield (pf/ft)
18	6.64	74	148	63	156
16	4.18	86	172	87	180

Product Features

- UL listed as Marine Shipboard Cable (File E86139)
- UL listed as Type MV-105 kV and 8 kV (File E66901)
- Cables pass UL 1685 and IEEE 383 vertical tray fire tests at 70,000 BTU/hr, IEC 332-3 category a fire test, IEEE 1202 and CSA FT4
- Cables are American Bureau of Shipping (ABS) listed as CWC MC Type MC Type MV/MC
- Cables are marked "-40°C" and are suitable for handling and installation down to -10°C (based on -40°C impact and bend tests per UL 1569)
- 105°C Temperature Rating
- Continuous, impervious metallic sheath corrugated for flexibility, prevents ingress of moisture, gases and liquids
- Aluminum armor resistance exceeds requirements of the NEC 2008 and NEC 2011 Article 250.178 for equipment grounding conductor
- Excellent mechanical & physical properties
- Minimal noise and signal interference
- Sunlight resistant jacket
- Suitable for direct burial and use in cable tray and embedment in concrete
- Suitable for Class I, II and III, Divisions 1 & 2 as per NEC 2008 and NEC 2011 (HL Rated)
- Three symmetrical grounding conductors for PW M/VFD and other modern AC drive/motor applications

Selling information

Options

The following constructions can be provided on special orders:

- Different conductor size
- Different pair or triad configurations
- Specially colored jackets
- Other constructions and combinations (some manufacturing restrictions apply)
  - UL 1309 listing and marking

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CORFLEX® MC

Armored Power and Control Cable

Part Number: PC

Armored Power and Control Cable UL Type MC HL, 600 V, 90°C rated - LEAD FREE

Description



Multi-conductor 14 AWG to 10 AWG:Multiple conductors and composites, with ground wire(s), continuous corrugated aluminum sheath, PVC jacket.

Construction

Conductor: Bare, annealed copper conforming to ASTM B3 and Class B stranded in accordance with ASTM B8.

Insulation: Cross linked polyethylene type XHHW-2 per UL 44.

Assembly: Conductors are cabled in concentric layers with or without grounding wire(s), interstices are filled with suitable non-hygroscopic fillers, as required. A binder tape of synthetic material assembles the core in an essentially round configuration.

Armor: Continuous corrugated aluminum sheath with no more than 0.2% trace copper providing complete protection against liquid & gas ingress. It also provides excellent mechanical protection, additional electrostatic shielding, and serves as an easy means of grounding equipment.

Jacket: Overall black polyvinyl chloride jacket per UL 1569, 90ºC temperature rating; low gas emission; limited flame spread and excellent corrosion resistance.

Conductor Identification - see link below

Bending Radius

Fixed position: 7 x cable overall diameter
During pulling: 12 x cable overall diameter

Specifications

- Meets UL 44, XHHW-2 600 V conductors
- Meets UL 1569 requirements for Type MC, Metal Clad cables
- Meets UL 2225 for Hazardous Locations
- Designated Type MC as per NEC Article 330
- Meets CSA C22.2 No. 123-08 for Aluminum Sheathed Cables
- Meets CSA C22.2 No. 174-M1984 for Hazardous Locations

Product features

Standards

National UL 1569;UL 2225;UL 44

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- UL approved cables Type MC, 600 V; File No. E47409
- UL approved insulated conductors
- Cables pass UL 1685 and IEEE 383 vertical tray fire tests at 70,000 BTU/hr, ICEA T-29-520 fire test at 210,000 BTU/hr, IEC 332-3 category A fire test, IEEE 1202 and CSA FT4
- Cables are American Bureau of Shipping (ABS) listed as CWC MC Type MC
- Cables exhibit a low temperature rating of -40ºC impact and –40ºC bend with suitable precautions
- Temperature rating of 90ºC dry and wet
- 130ºC emergency rating & 250ºC short circuit rating
- Continuous, impervious metallic sheath corrugated for flexibility, prevents ingress of moisture, gases and liquids
- Aluminum sheath cross-section exceeds requirements of the NEC Section 250.122 for grounding conductor
- Sheath provides good electronic shielding so that Corflex® can be used in certain instrumentation applications when adequately grounded
- Excellent mechanical & physical properties
- Sunlight resistant jacket
- Suitable for direct burial and use in cable tray and embedment in concrete
- LEAD FREE

Characteristics

Construction characteristics	
Conductor material	Copper
Electrical characteristics	
Maximum operating voltage	600 V
Usage characteristics	
Maximum operating temperature	90 °C

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Multiconductors, with Bare Gound(s) Electrical Data

Part Number	# of Cond.	Cond. Size AWG/ kcmil	DC Resistance		AC Resistance 90°C, 60 Hz Ω/kft	Inductive Reactance (Ω/kft@ 60Hz)	Voltage Drop V/(A.Kft)	Ampacities Note 1	
			20°C Ω/kft	25°C Ω/kft				75°C	90°C
*670155	2	14(7w)	2.5553	2.6064	3.2583	0.0376	2.9489	15	15
*670142	3	14(7w)	2.5553	2.6064	3.2583	0.0376	2.9489	15	15
*317677	4	14(7w)	2.5553	2.6064	3.2583	0.0376	2.9489	15	15
*318345	5	14(7w)	2.5553	2.6064	3.2583	0.0497	2.9542	15	15
*318352	7	14(7w)	2.5553	2.6064	3.2583	0.0545	2.9566	14	15
*318360	9	14(7w)	2.5553	2.6064	3.2583	0.0596	2.9585	14	15
*318378	12	14(7w)	2.5553	2.6064	3.2583	0.0641	2.9604	10	13
318386	15	14(7w)	2.5553	2.6064	3.2583	0.0666	2.9615	10	13
*318394	19	14(7w)	2.5553	2.6064	3.2583	0.0694	2.9627	10	13
*318402	25	14(7w)	2.5553	2.6064	3.2583	0.0743	2.9649	9	11
*318410	37	14(7w)	2.5553	2.6064	3.2583	0.0787	2.9668	8	10
*670156	2	12(7w)	1.6082	1.6404	2.0506	0.0353	1.8610	20	20
*670144	3	12(7w)	1.6082	1.6404	2.0506	0.0353	1.8610	20	20
*317693	4	12(7w)	1.6082	1.6404	2.0506	0.0353	1.8610	20	20
*318436	5	12(7w)	1.6082	1.6404	2.0506	0.0475	1.8663	20	20
*318444	7	12(7w)	1.6082	1.6404	2.0506	0.0526	1.8685	18	20
*318451	9	12(7w)	1.6082	1.6404	2.0506	0.0574	1.8706	18	20
*318469	12	12(7w)	1.6082	1.6404	2.0506	0.0620	1.8726	13	15
318447	15	12(7w)	1.6082	1.6404	2.0506	0.0644	1.8736	13	15
318485	19	12(7w)	1.6082	1.6404	2.0506	0.0672	1.8749	13	15
318493	25	12(7w)	1.6082	1.6404	2.0506	0.0719	1.8769	11	14
318501	37	12(7w)	1.6082	1.6404	2.0506	0.0764	1.8789	10	12
670157	2	10(7w)	1.0118	1.2902	1.2902	0.0332	1.1756	30	30
*670146	3	10(7w)	1.0118	1.2902	1.2902	0.0332	1.1756	30	30
*317719	4	10(7w)	1.0118	1.2902	1.2902	0.0332	1.1756	28	30
318527	5	10(7w)	1.0118	1.2902	1.2902	0.0454	1.1809	28	30
318535	7	10(7w)	1.0118	1.2902	1.2902	0.0507	1.1832	25	28
318543	9	10(7w)	1.0118	1.2902	1.2902	0.0579	1.1864	25	28
*318550	12	10(7w)	1.0118	1.2902	1.2902	0.0601	1.1874	18	20
665148	37	10(7w)	1.0118	1.2902	1.2902	0.0744	1.1936	14	16

* Stock Item

Notes:
1) Ampacities are in accordance with NEC 2008 Table 310.16 or NEC 2011 Table 310.15(B)(16) for conductors in raceway or direct buried at 30°C ambiant temperature and 90°C conductor temperature. The overcurrent protection shall not exceed 15 amperes for 14 AWG, 20 amperes for 12 AWG, and 30 amperes for 10 AWG copper conductors after any correction factors for ambient temperature and number of conductors have been applied (NEC 2008 and NEC 2011 Article 240.4(D). For correction factors for different ambient temperatures and ampacities at different conductor temperatures, see NEC 2008 Table 310.16 or NEC 2011 Table 310.15(B)(16). Ampacities for cables having more than three conductors have been derated per NEC 2008 Article 310.15(B)(2)(a) or NEC 2011 Article 310.15(B)(3)(a).

Multiconductors, with Bare Gound(s) Physical Data

Part Number	# of Cond.	Cond. Size AWG/ kcmil	Insulation Thickness mils	Ground Wire Size AWG	Nominal Diameter over Core(in)	Nominal Diameter over Sheath(in)	Jacket Thickness mils	Nominal Diameter over Jacket(in)	Approx. Net Cable Weight (lb/kft)
*670155	2	14(7w)	30	14(7w)	0.273	0.469	50	0.581	155
*670142	3	14(7w)	30	3x18(7w)	0.390	0.555	50	0.660	200
*317677	4	14(7w)	30	14(7w)	0.336	0.503	50	0.606	191
*318345	5	14(7w)	30	14(7w)	0.366	0.532	50	0.635	212
*318352	7	14(7w)	30	14(7w)	0.417	0.601	50	0.704	263
*318360	9	14(7w)	30	14(7w)	0.486	0.645	50	0.748	307
*318378	12	14(7w)	30	14(7w)	0.560	0.783	50	0.887	388
318386	15	14(7w)	30	14(7w)	0.610	0.811	50	0.915	443
*318394	19	14(7w)	30	14(7w)	0.669	0.921	50	1.028	572
*318402	25	14(7w)	30	14(7w)	0.797	1.005	50	1.111	691
*318410	37	14(7w)	30	14(7w)	0.933	1.218	50	1.323	986
*670156	2	12(7w)	30	12(7w)	0.345	0.512	50	0.614	185
*670144	3	12(7w)	30	3x16(7w)	0.340	0.555	50	0.660	226
*317693	4	12(7w)	30	12(7w)	0.385	0.550	50	0.653	239
*318436	5	12(7w)	30	12(7w)	0.414	0.599	50	0.702	280
*318444	7	12(7w)	30	12(7w)	0.478	0.640	50	0.744	338
*318451	9	12(7w)	30	12(7w)	0.551	0.777	50	0.881	437
*318469	12	12(7w)	30	12(7w)	0.639	0.828	50	0.932	502
318447	15	12(7w)	30	12(7w)	0.691	0.936	50	1.039	635
318485	19	12(7w)	30	12(7w)	0.760	0.981	50	1.085	743
318493	25	12(7w)	30	12(7w)	0.897	1.190	50	1.295	987
318501	37	12(7w)	30	12(7w)	1.059	1.374	50	1.478	1365
670157	2	10(7w)	30	10(7w)	0.360	0.527	50	0.632	231
*670146	3	10(7w)	30	3x14(7w)	0.450	0.620	50	0.725	312
*317719	4	10(7w)	30	10(7w)	0.448	0.621	50	0.724	319
318527	5	10(7w)	30	10(7w)	0.479	0.641	50	0.743	364
318535	7	10(7w)	30	10(7w)	0.556	0.780	50	0.884	484
318543	9	10(7w)	30	10(7w)	0.693	0.936	50	1.040	630
*318550	12	10(7w)	30	10(7w)	0.744	0.971	50	1.074	732
665148	37	10(7w)	30	10(7w)	1.227	1.591	60	1.724	1984
*Stock Item									

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Composite Electrical Data

Part Number	# of Cond.	Cond. Size AWG/ kcmil	DC Resistance		AC Resistance 90ºC, 60 Hz Ω/kft	Inductive Reactance (Ω/kft@ 60Hz)	Voltage Drop V/(A.Kft)	Ampacities Note 1	
			20ºC Ω/kft	25ºC Ω/kft				75ºC	90ºC
	(4)	(3)							
199760	12(7w)	10(7w)	1.0118	1.0320	1.2902	0.0332	1.1756	30	30
*290783	12(7w)	8(7w)	0.6361	0.6488	0.8111	0.0348	0.7452	50	55
290791	12(7w)	6(7w)	0.4002	0.4082	0.5104	0.0328	0.4737	65	75
290809	12(7w)	4(7w)	0.2516	0.2566	0.3209	0.0312	0.3024	85	95
654277	12(7w)	2(7w)	0.1574	0.1605	0.2009	0.0299	0.1938	115	130
Composite - 3 POWER Conductors with 3X12 AWG CONTROL Conductors and 1 Bare Ground									
	(3)	(3)							
665024	12(7w)	10(7w)	1.0118	1.0320	1.2902	0.0332	1.1756	30	30
665025	12(7w)	8(7w)	0.6361	0.6488	0.8111	0.0348	0.7452	50	55
665027	12(7w)	6(7w)	0.4002	0.4082	0.5104	0.0328	0.4737	65	75
665028	12(7w)	4(7w)	0.2516	0.2566	0.3209	0.0312	0.3024	85	95
665035	12(7w)	2(7w)	0.1574	0.1605	0.2009	0.0299	0.1938	115	130
Composite - 4 POWER Conductors with 4X12 AWG CONTROL Conductors and 1 Bare Ground									
	(4)	(4)							
665019	12(7w)	10(7w)	1.0118	1.0320	1.2902	0.0332	1.1756	30	30
665020	12(7w)	8(7w)	0.6361	0.6488	0.8111	0.0348	0.7452	50	55
665021	12(7w)	6(7w)	0.4002	0.4082	0.5104	0.0328	0.4737	65	75
665022	12(7w)	4(7w)	0.2516	0.2566	0.3209	0.0312	0.3024	85	95
665023	12(7w)	2(7w)	0.1574	0.1605	0.2009	0.0299	0.1938	115	130
Composite - 3 POWER Conductors with 4 X14 AWG CONTROL Conductors and 1 Bare Ground									
	(4)	(3)							
665036	14(7w)	10(7w)	1.0118	1.0320	1.2902	0.0332	1.1756	30	30
665037	14(7w)	8(7w)	0.6361	0.6488	0.8111	0.0348	0.7452	50	55
665038	14(7w)	6(7w)	0.4002	0.4082	0.5104	0.0328	0.4737	65	75
665039	14(7w)	4(7w)	0.2516	0.2566	0.3209	0.0312	0.3024	85	95
665040	14(7w)	2(7w)	0.1574	0.1605	0.2009	0.0299	0.1938	115	130
Composite - 3 POWER Conductors with 3 X14 AWG CONTROL Conductors and 1 Bare Ground									
	(3)	(3)							
665041	14(7w)	10(7w)	1.0118	1.0320	1.2902	0.0332	1.1756	30	30
665042	14(7w)	8(7w)	0.6361	0.6488	0.8111	0.0348	0.7452	50	55
665044	14(7w)	6(7w)	0.4002	0.4082	0.5104	0.0328	0.4737	65	75
665045	14(7w)	4(7w)	0.2516	0.2566	0.3209	0.0312	0.3024	85	95
665046	14(7w)	2(7w)	0.1574	0.1605	0.2009	0.0299	0.1938	115	130
Composite - 4 POWER Conductors with 4 X14 AWG CONTROL Conductors and 1 Bare Ground									
	(4)	(4)							
665047	14(7w)	10(7w)	1.0118	1.0320	1.2902	0.0332	1.1756	30	30
665048	14(7w)	8(7w)	0.6361	0.6488	0.8111	0.0348	0.7452	50	55
665049	14(7w)	6(7w)	0.4002	0.4082	0.5104	0.0328	0.4737	65	75
665050	14(7w)	4(7w)	0.2516	0.2566	0.3209	0.0312	0.3024	85	95
665051	14(7w)	2(7w)	0.1574	0.1605	0.2009	0.0299	0.1938	115	130

* Stock item

All drawings, designs, specifications, plans and particulars of weights, size and dimensions contained in the technical or commercial documentation of Nexans is indicative only and shall not be binding on Nexans or be treated as constituting a representation on the part of Nexans.

CORFLEX® MC

Armored Power and Control Cable

Composite Physical Data

Part Number	CONTROL Size AWG	Insulation Thickness mils	POWER Size AWG	Insulation Thickness mils	Ground Wire Size AWG	Nominal Diameter over Core(in)	Nominal Diameter over Sheath(in)	Jacket Thickness mils	Nominal Diameter over Jacket(in)	Approx. Net Cable Weight (lbs/kft)
Composite - 3 POWER Conductors with 4X12 AWG CONTROL Conductors and 1 Bare Ground										
	(4)		(3)							
199760	12(7w)	30	10(7w)	30	10(7w)	0.540	0.772	50	0.876	425
*290783	12(7w)	30	8(7w)	45	10(7w)	0.664	0.917	50	1.026	565
290791	12(7w)	30	6(7w)	45	8(7w)	0.750	0.975	50	1.077	685
290809	12(7w)	30	4(7w)	45	8(7w)	0.845	1.149	50	1.254	926
654277	12(7w)	30	2(7w)	45	6(7w)	0.870	1.168	50	1.273	1198
Composite - 3 POWER Conductors with 3X12 AWG CONTROL Conductors and 1 Bare Ground										
	(3)		(3)							
665024	12(7w)	30	10(7w)	30	10(7w)	0.494	0.750	50	0.854	396
665025	12(7w)	30	8(7w)	45	10(7w)	0.590	0.800	50	0.904	473
665027	12(7w)	30	6(7w)	45	8(7w)	0.695	0.937	50	1.041	665
665028	12(7w)	30	4(7w)	45	8(7w)	0.735	0.964	50	1.068	807
665035	12(7w)	30	2(7w)	45	6(7w)	0.860	1.160	50	1.265	1170
Composite - 4 POWER Conductors with 4X12 AWG CONTROL Conductors and 1 Bare Ground										
	(4)		(4)							
665019	12(7w)	30	10(7w)	30	10(7w)	0.560	0.782	50	0.886	463
665020	12(7w)	30	8(7w)	45	10(7w)	0.659	0.914	50	1.024	630
665021	12(7w)	30	6(7w)	45	8(7w)	0.725	0.958	50	1.062	778
665022	12(7w)	30	4(7w)	45	8(7w)	0.807	1.012	50	1.117	993
665023	12(7w)	30	2(7w)	45	6(7w)	0.927	1.214	50	1.319	1436
Composite - 3 POWER Conductors with 4 X14 AWG CONTROL Conductors and 1 Bare Ground										
	(4)		(3)							
665036	14(7w)	30	10(7w)	30	10(7w)	0.510	0.754	50	0.860	387
665037	14(7w)	30	8(7w)	45	10(7w)	0.610	0.811	50	0.915	478
665038	14(7w)	30	6(7w)	45	8(7w)	0.700	0.941	50	1.045	640
665039	14(7w)	30	4(7w)	45	8(7w)	0.730	0.961	50	1.064	799
665040	14(7w)	30	2(7w)	45	6(7w)	0.820	1.128	50	1.240	1165
Composite - 3 POWER Conductors with 3 X14 AWG CONTROL Conductors and 1 Bare Ground										
	(3)		(3)							
665041	14(7w)	30	10(7w)	30	10(7w)	0.470	0.635	50	0.738	340
665042	14(7w)	30	8(7w)	45	10(7w)	0.591	0.800	50	0.904	457
665044	14(7w)	30	6(7w)	45	8(7w)	0.670	0.921	50	1.028	641
665045	14(7w)	30	4(7w)	45	8(7w)	0.770	0.987	50	1.091	788
665046	14(7w)	30	2(7w)	45	6(7w)	0.885	1.180	50	1.285	1146
Composite - 4 POWER Conductors with 4 X14 AWG CONTROL Conductors and 1 Bare Ground										
	(4)		(4)							
665047	14(7w)	30	10(7w)	30	10(7w)	0.550	0.776	50	0.880	428
665048	14(7w)	30	8(7w)	45	10(7w)	0.645	0.831	50	0.935	535
665049	14(7w)	30	6(7w)	45	8(7w)	0.695	0.938	50	1.042	749
665050	14(7w)	30	4(7w)	45	8(7w)	0.787	0.999	50	1.103	953
665051	14(7w)	30	2(7w)	45	6(7w)	0.927	1.214	50	1.319	1401

*Stock Item

3 Conductors with 3 Bare Grounds* Electrical Data

Part Number	Cond. Size AWG/ kcmil	Ground Wire Size AWG	DC Resistance		AC Resistance 90ºC, 60 Hz Ω/kft	Inductive Reactance (Ω/kft @60Hz)	Voltage Drop V/(A.Kft)	Ampacities Note 1	
			20ºC Ω/kft	25ºC Ω/kft				75ºC	90ºC
670142	14(7w)	3x18(7w)	2.5553	2.6064	3.2583	0.0376	2.9489	15	15
670144	12(7w)	3x16(7w)	1.6082	1.6404	2.0506	0.0353	1.8610	20	20
670146	10(7w)	3x14(7w)	1.0118	1.0320	1.2902	0.0332	1.1756	30	30
*670148	8(7w)	3x14(7w)	0.6361	0.6488	0.8111	0.0348	0.7452	50	55
*670149	6(7w)	3x12(7w)	0.4002	0.4082	0.5104	0.0329	0.4737	65	75
*670150	4(7w)	3x12(7w)	0.2516	0.2566	0.3209	0.0312	0.3024	85	95
*670151	2(7w)	3x10(7w)	0.1574	0.1605	0.2009	0.0299	0.1938	115	130
670152	1(19w)	3x10(7w)	0.1255	0.1280	0.1603	0.0288	0.1568	130	145
*670153	1/0(19w)	3x10(7w)	0.0999	0.1019	0.1278	0.0281	0.1272	150	170
*670118	2/0(19w)	3x10(7w)	0.0797	0.0813	0.1021	0.0280	0.1041	175	195
*670119	3/0(19w)	3x8(7w)	0.0629	0.0642	0.0808	0.0275	0.0847	200	225
*670120	4/0(19w)	3x8(7w)	0.0497	0.0507	0.0641	0.0271	0.0695	230	260
*670121	250(37w)	3x8(7w)	0.0424	0.0432	0.0584	0.0263	0.0608	255	290
*670122	350(37w)	3x6(7w)	0.0301	0.0307	0.0395	0.0263	0.0470	310	350
*670123	500(37w)	3x6(7w)	0.0212	0.0216	0.0290	0.0250	0.0367	380	430

*Stock Item

Notes:
1) Ampacities are based on NEC 2011 Table 310.15(B)(16) for not more than three current-carrying conductors in raceway, cable, or earth (direct buried), based on an ambient temperature of 30°C (86°F). Refer to Table 310.15(B)(2) for the ampacity correction factors where the ambient temperature is other than 30°C (86°F).
2) Three conductor cables with 3 ground wires are also excellent for use with variable frequency drives. In addition to UL, these 3-conductor constructions are also certified to CSA C22.2 No. 123 and CSA C22.2 No. 174.

All drawings, designs, specifications, plans and particulars of weights, size and dimensions contained in the technical or commercial documentation of Nexans is indicative only and shall not be binding on Nexans or be treated as constituting a representation on the part of Nexans.

3 Conductors with 3 Bare Grounds* Physical Data

Part Number	Cond. Size AWG/ kcmil	Insulation Thickness mils	Ground Wire Size AWG	Nominal Diameter over Core(in)	Nominal Diameter over Sheath(in)	Jacket Thickness mils	Nominal Diameter over Jacket(in)	Approx. Net Cable Weight (lb/kft)
670142	14(7w)	30	3x18(7w)	0.390	0.555	50	0.660	200
670144	12(7w)	30	3x16(7w)	0.340	0.555	50	0.660	226
670146	10(7w)	30	3x14(7w)	0.450	0.620	50	0.725	312
*670148	8(7w)	45	3x14(7w)	0.520	0.750	50	0.838	413
*670149	6(7w)	45	3x12(7w)	0.600	0.802	50	0.905	542
*670150	4(7w)	45	3x12(7w)	0.700	0.937	50	1.039	735
*670151	2(7w)	45	3x10(7w)	0.830	1.127	50	1.232	1097
670152	1(19w)	55	3x10(7w)	0.950	1.230	50	1.320	1330
*670153	1/0(19w)	55	3x10(7w)	1.040	1.350	50	1.473	1592
*670118	2/0(19w)	55	3x10(7w)	1.126	1.422	50	1.510	1882
*670119	3/0(19w)	55	3x8(7w)	1.250	1.606	60	1.739	2400
*670120	4/0(19w)	55	3x8(7w)	1.360	1.734	60	1.867	2910
*670121	250(37w)	65	3x8(7w)	1.477	1.925	60	2.058	3316
*670122	350(37w)	65	3x6(7w)	1.685	2.028	60	2.162	4375
*670123	500(37w)	65	3x6(7w)	1.954	2.340	75	2.504	6026

*Stock Item

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4 Conductors with 1 Bare Ground Electrical Data

Part Number	Cond. Size AWG/ kcmil	Ground Wire Size AWG	DC Resistance		AC Resistance 90ºC, 60 Hz Ω/kft	Inductive Reactance (Ω/kft @60Hz)	Voltage Drop V/(A.Kft)	Ampacities Note 1	
			20ºC Ω/kft	25ºC Ω/kft				75ºC	90ºC
317677	14(7w)	14(7w)	2.5553	2.6064	3.2583	0.0375	2.9307	15	15
317693	12(7w)	12(7w)	1.6082	1.6404	2.0506	0.0353	1.8510	20	20
317719	10(7w)	10(7w)	1.0118	1.0320	1.2902	0.0332	1.1758	30	30
*317735	8(7w)	10(7w)	0.6361	0.6488	0.8111	0.0348	0.7472	50	55
*317750	6(7w)	8(7w)	0.4002	0.4082	0.5104	0.0329	0.4737	65	75
*317776	4(7w)	8(7w)	0.2516	0.2566	0.3209	0.0312	0.3024	85	95
*317792	2(7w)	6(7w)	0.1574	0.1605	0.2009	0.0299	0.1938	115	130
698571	1(19w)	6(7w)	0.1255	0.1280	0.1603	0.0288	0.1567	130	145
317818	1/0(19w)	6(7w)	0.0999	0.1019	0.1278	0.0283	0.1268	150	170
*317834	2/0(19w)	6(7w)	0.0790	0.0806	0.1011	0.0281	0.1033	175	195
660631	3/0(19w)	4(7w)	0.0627	0.0640	0.0804	0.0276	0.0844	200	225
*317867	4/0(19w)	4(7w)	0.0497	0.0507	0.0641	0.0271	0.0694	230	260
670577	250(37w)	4(7w)	0.0424	0.0432	0.0584	0.0264	0.0607	255	290
*317891	350(37w)	3(7w)	0.0301	0.0307	0.0395	0.0263	0.0468	310	350
*317917	500(37w)	2(7w)	0.0212	0.0216	0.0282	0.0250	0.0363	380	430

*Stock Item

Notes:
1) Ampacities are based on NEC 2011 Table 310.15(B)(16) for not more than three current-carrying conductors (where the 4th conductor is the neutral of a 3-phase, 4 wire system) in raceway, cable, or earth (direct buried), based on an ambient temperature of 30°C (86°F). Refer to Table 310.15(B)(2) for the ampacity correction factors where the ambient temperature is other than 30°C (86°F).

All drawings, designs, specifications, plans and particulars of weights, size and dimensions contained in the technical or commercial documentation of Nexans is indicative only and shall not be binding on Nexans or be treated as constituting a representation on the part of Nexans.

4 Conductors with 1 Bare Ground Physical Data

Part Number	Cond. Size AWG/ kcmil	Insulation Thickness mils	Ground Wire Size AWG	Nominal Diameter over Core(in)	Nominal Diameter over Sheath(in)	Jacket Thickness mils	Nominal Diameter over Jacket(in)	Approx. Net Cable Weight (lb/kft)
317677	14(7w)	30	14(7w)	0.336	0.503	50	0.606	191
317693	12(7w)	30	12(7w)	0.396	0.549	50	0.652	245
317719	10(7w)	30	10(7w)	0.450	0.620	50	0.730	320
*317735	8(7w)	45	10(7w)	0.585	0.795	50	0.900	465
*317750	6(7w)	45	8(7w)	0.680	0.930	50	1.027	675
*317776	4(7w)	45	8(7w)	0.782	0.995	50	1.095	892
*317792	2(7w)	45	6(7w)	0.927	1.214	50	1.320	1332
698571	1(19w)	55	6(7w)	1.041	1.361	50	1.466	1628
317818	1/0(19w)	55	6(7w)	1.134	1.427	50	1.525	1922
*317834	2/0(19w)	55	6(7w)	1.238	1.598	60	1.732	2393
660631	3/0(19w)	55	4(7w)	1.356	1.730	60	1.865	2975
*317867	4/0(19w)	55	4(7w)	1.500	1.936	60	2.069	3605
670577	250(37w)	65	4(7w)	1.653	2.012	60	2.147	4110
*317891	350(37w)	65	3(7w)	1.889	2.230	75	2.387	5617
*317917	500(37w)	65	2(7w)	2.196	2.694	75	2.861	7892
*Stock Item								

CONDUCTOR IDENTIFICATION

Power and control:
Multi-conductor 14 AWG to 10 AWG:
Method #1-E2 per ICEA S-73-532
3 conductor 8 AWG to 2 AWG:
Method #1-E2 per ICEA S-73-532,
3 conductor 1 AWG to 500 kcmil:
Method 4 per ICEA S-73-532
2 & 4 conductor 8 AWG to 750 kcmil:
Method #4 per ICEA S-73-532

Composite power and control:
Power conductors:
Method #4 per ICEA S-73-532
Control conductors:
Method #1-E2 per ICEA S-73-532

Selling information

Options

The following constructions can be provided on special orders:

All drawings, designs, specifications, plans and particulars of weights, size and dimensions contained in the technical or commercial documentation of Nexans is indicative only and shall not be binding on Nexans or be treated as constituting a representation on the part of Nexans.

- Aluminum conductors
- Extra ground wires
- Special color or number coding
- Specially colored jackets
- Other constructions and combinations (some manufacturing restrictions apply)
  - UL 1309 listing and marking

All drawings, designs, specifications, plans and particulars of weights, size and dimensions contained in the technical or commercial documentation of Nexans is indicative only and shall not be binding on Nexans or be treated as constituting a representation on the part of Nexans.

CORFLEX® MC

CORFLEX® PL

Part Number: PL

Armored Instrumentation Cable UL Type PLTC & ITC, 300 V, 105°C rated - LEAD FREE

Description



Application

Single or multiple individually shielded pairs or triads, overall cable shield, PVC inner jacket, continuous corrugated and welded, impervious aluminum sheath, PVC jacket

Construction

Conductor: Bare, annealed copper conforming to ASTM B3 and Class B stranded in accordance with ASTM B8.

Insulation: Polyvinyl chloride in accordance with UL 13 and 2250, flame retardant, 105°C temperature rating.

Insulation shield: Aluminum foil/polyester shield helically wrapped to provide 100% coverage and tinned copper drain wire that is two gauge sizes smaller than the circuit conductors. These shields are electrically isolated from each other.

Assembly: Pairs/triads are cabled in concentric layers with interstices filled with suitable non-hygroscopic fillers, as required. A binder tape of synthetic material assembles the core in an essentially round configuration.

Overall cable shield: Aluminum foil/polyester shield helically wrapped to provide 100% coverage and tinned copper drain wire that is the same size as the circuit conductors, with the exception of single pair/triad constructions where the drain wire is two gauge sizes smaller than the circuit conductors.

Inner jacket: Polyvinyl chloride jacket, over cabled core as per UL 13 and UL 2250, 90°C temperature rating, with additional resistance to fire spread. A rip cord is laid longitudinally under the jacket to facilitate stripping.

Armor: Continuous corrugated and welded, impervious aluminum sheath with no more than 0.2% trace copper providing complete protection against liquid and gas ingress. It also provides excellent mechanical protection, additional electrostatic shielding, and serves as an easy means of grounding equipment.

Jacket: Overall black polyvinyl chloride jacket per UL 13 and UL 2250, 90°C temperature rating; low acid gas emission; limited flame spread and excellent corrosion resistance.

Conductor Identification

Pairs: black/white & number coded

Triads: black/white/red & number coded

Bending Radius

Fixed position: 7 x cable overall diameter

During pulling: 12 x cable overall diameter

Specifications

* Meets UL 13 and UL 2250, rated 105°C 300 V conductor
* Meets UL requirements for Type PLTC, power limited circuit cable and ITC, instrumentation tray cable
* Designated Type PLTC,Power Limited Tray Cable, per NEC 2008 NEC 2011Article and ITC, Instrumentation Tray Cable, per NEC 2008 NEC 2011Article.

* Cables are American Bureau of Shipping (ABS) listed as CWC MC Type PLTC.

Standards

National UL 13;UL 2250

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Characteristics

Construction characteristics	
Conductor material	Aluminium
Electrical characteristics	
Maximum operating voltage	300 V
Usage characteristics	
Maximum operating temperature	105 °C

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Multi Pairs, 300 V - 18 AWG (7w) SPOS

Part Number	# of Pairs	PVC Insulation Thickness (mils)	Nominal Diameter over Core (inches)	Inner Jacket Thickness (mils)	Nominal Diameter over Inner Jacket (inches)	Nominal Diameter over Sheath (inches)	Outer Jacket Thickness (mils)	Nominal Diameter over Outer Jacket (inches)	Approximate Net Cable Weight (lb/kft)
665101	2	15	0.277	50	0.363	0.529	50	0.632	225
665102	4	15	0.329	50	0.435	0.613	50	0.715	250
*068718	8	15	0.421	50	0.527	0.763	50	0.867	430
665103	12	15	0.562	60	0.688	0.933	50	1.037	565
665104	16	15	0.634	60	0.760	0.981	50	1.085	735
665105	20	15	0.699	70	0.824	1.132	50	1.241	760
665106	24	15	0.782	70	0.929	1.215	50	1.320	1040
665107	36	15	0.941	70	1.084	1.392	50	1.492	1108
* Stock items									

Multi Pairs, 300 V - 16 AWG (7w) SPOS

Part Number	# of Pairs	PVC Insulation Thickness (mils)	Nominal Diameter over Core (inches)	Inner Jacket Thickness (mils)	Nominal Diameter over Inner Jacket (inches)	Nominal Diameter Over Sheath (inches)	Outer Jacket Thickness (mils)	Nominal Diameter over Outer Jacket (inches)	Approximate Net Cable Weight (lb/kft)
*066555	1	15	0.185	35	0.259	0.460	50	0.566	125
*066571	2	15	0.315	50	0.421	0.604	50	0.706	260
*066597	4	15	0.376	50	0.502	0.749	50	0.856	355
*066605	8	15	0.504	60	0.630	0.822	50	0.926	555
*066613	12	15	0.644	60	0.767	0.985	50	1.086	770
665108	16	15	0.727	70	0.853	1.155	50	1.260	816
665109	20	15	0.801	70	0.947	1.230	50	1.335	1150
*066647	24	15	0.897	70	1.043	1.363	50	1.467	1180
665110	36	15	1.079	80	1.227	1.591	60	1.725	1850
*Stock items									

Multi Triads, 300 V - 16 AWG (7w) STOS

Part Number	# of Pairs	PVC Insulation Thickness (mils)	Nominal Diameter over Core (inches)	Inner Jacket Thickness (mils)	Nominal Diameter over Inner Jacket (inches)	Nominal Diameter over Sheath (inches)	Outer Jacket Thickness (mils)	Nominal Diameter over Outer Jacket (inches)	Approximate Net Cable Weight (lb/kft)
665111	1	15	.199	35	.275	.460	50	.567	140
*083410	4	15	.428	50	.533	.789	50	.894	400
665112	8	15	.605	60	.730	.961	50	1.284	617
665113	12	15	.785	70	.896	1.181	50	1.348	905
665114	16	15	.834	70	.981	1.370	50	1.478	1200
665115	24	15	1.078	70	1.226	1.590	90	1.786	1632
*Stock items									

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Electrical Properties

300 V Shielded Pairs / Triads with an overall Cable Shield

Conductor Size (AWG)	DC Resistance 20°C Ω/kft	Capacitance			
		Pairs		Triads	
		Conductor -Conductor (pf/ft)	Conductor -Shield (pf/ft)	Conductor -Conductor (pf/ft)	Conductor -Shield (pf/ft)
18	6.64	61	122	64	125
16	4.18	72	143	76	147

Selling information

Options

The following constructions can be provided on special orders:

- Different conductor size
- Different pair or triad configurartions
- Specially colored jackets
- Other constructions and combinations (some manufacturing restrictions apply)
- UL 1309 listing and marking

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